

Amit Agarwal

Applied Scientist & Technical Lead

Retrieval, Agentic AI, Multimodal Models & Evaluation

Seattle, WA | amit.pinaki@gmail.com | +1-510-862-4544

linkedin.com/in/amitagarwal6 | github.com/amitbcp | iamamitagarwal.github.io | Google Scholar

Professional Summary

Applied scientist and technical lead with **10+ years** in enterprise AI, spanning search, retrieval, agents, and multimodal models. Leads work from model development and architecture through evaluation and product integration. Current focus: shared knowledge infrastructure, multimodal search, and agent evaluation harnesses.

- **Enterprise delivery:** Led agentic and knowledge systems across **15+ engagements**; delivered document AI capabilities to **20+ customers**.
- **Retrieval research:** Co-first author of enterprise reranking research reporting **0.64 MRR@10 vs. 0.57** for ADORE+STAR on a cloud-services test set (ACL 2025 Industry).
- **Research recognition:** First-author work on multilingual alignment and multimodal evaluation; co-author of *AccessEval*, recipient of the **EMNLP 2025 Social Impact Award**.

Experience

Oracle Cloud Infrastructure (OCI)

Apr 2021 - Present

Principal Applied Scientist (current), Generative AI

Seattle, WA

- Architected a shared knowledge store for retrieval-augmented generation (RAG) and agents, combining knowledge graphs, multimodal and multilingual retrieval, reranking, and access-aware grounding.
- Led retrieval and ranking across SQL/code, documents, images, audio, and video; trained in-house multimodal and multilingual embedding and ranking models.
- Implemented streaming multimodal evaluation with checks for duplicate query IDs and controlled retrieval runs; developing REST/MCP interfaces for governed context delivery.
- Led a multi-agent platform for RAG, natural-language-to-SQL (NL2SQL), code, and enterprise workflows across **15+ engagements**; established tool-use, customer-feedback, and evaluation loops.
- Co-owned multimodal RAG science for Oracle Fusion, contributing to an image-to-text capability deployed to production in August 2025.
- Delivered optical character recognition (OCR), visual question answering (VQA), layout understanding, and key information extraction (KIE) capabilities to **20+ customers**; advanced low-data document AI with graph neural networks.

Jio Platforms (Reliance)

Jun 2020 - Apr 2021

Data Science Lead, Computer Vision

Hyderabad, India

- Led computer-vision development from data and models through evaluation and product integration, covering detection, tracking, scene understanding, and security/compliance for video analytics.
- Built explainable medical imaging for Reliance Hospitals and OCR-based document understanding and information retrieval.

Abzooba Infotech (acquired by UST Global)

Jun 2016 - Jun 2020

Data Scientist

Pune, India

- Served as product owner and technical lead for financial-news recommendation and shipping-disclosure classification systems; developed query understanding, named entity recognition, and implicit-feedback models.
- Built document search, summarization, and image disambiguation, along with customer-feedback analytics and predictive workflows for a Singapore government SmartCity program.

Selected Publications

- **Hard Negative Mining for Domain-Specific Retrieval in Enterprise Systems**
ACL 2025 Industry. Co-first author (equal contribution). Hard-negative mining for enterprise rerankers; reported **MRR@10 of 0.64 vs. 0.57** for ADORE+STAR on a cloud-services test set.
- **Aligning LLMs for Multilingual Consistency in Enterprise Applications**
EMNLP 2025 Industry. First author. Batch-aligned preference optimization; Llama-3.1-70B achieved **74.9% average MGSM exact match vs. 57.2%** for the baseline in the reported study.
- **AccessEval: Benchmarking Disability Bias in Large Language Models**
EMNLP 2025 | Social Impact Award. Co-author. Evaluated **21 LLMs across 6 domains and 9 disability types**, using paired queries to measure changes in factual accuracy, sentiment, and social perception.
- **Do Image-Text Metrics Respect Semantic Invariances?**
ACL 2026 Findings. First author. Evaluated five image-text metrics under meaning-preserving changes; post-hoc calibration roughly halved median sensitivity in the reported experiments.
- **Pushing on Multilingual Reasoning Models with Language-Mixed Chain-of-Thought**
ICLR 2026. Co-author. Combined English and target-language reasoning in training and distillation, addressing multilingual consistency in long-chain reasoning.

Additional work spans ACL, EMNLP, NAACL, CVPR, ICLR, COLING, DASFAA, and CIKM. Full publication list: [Google Scholar](#).

Technical Skills

Search and agents: Retrieval-augmented generation (RAG), embeddings, reranking, knowledge graphs, multimodal and multilingual retrieval, agent orchestration, REST, Model Context Protocol (MCP).

Model development: Supervised fine-tuning (SFT), direct preference optimization (DPO), RLHF, reward modeling, distillation, quantization, synthetic data, document AI, graph neural networks.

Evaluation: Retrieval and agent evaluation harnesses, benchmark design, baseline comparisons, safety evaluation, image-text metrics.

Engineering: Python, PyTorch, TensorFlow, PyTorch Lightning, OCI, AWS, Azure, Docker, Git.

Education

MS, Machine Learning & Artificial Intelligence, Distinction, Liverpool John Moores University

BTech, Computer Science & Engineering, IEM, West Bengal University of Technology

Research Service & Open Collaboration

- **Workshop organizing:** GRAIL-V at CVPR 2026, SURGeLLM at ACL 2026, and DocInsights at EMNLP 2026.
- **Research service:** Area chair, program committee member, and reviewer across AAAI, NeurIPS, ACL, CVPR, EMNLP, NAACL, IJCAI, and ARR.
- **Mentoring and talks:** Speaker at MIPR 2024 and ICDMAI 2020; data-science mentor with Great Learning, MIT, and UT Austin programs.
- **Open source:** Contributions and collaborations across VLMEvalKit, EasyOCR, PyTorch Lightning, and Flair.

Awards

EMNLP 2025 Social Impact Award (AccessEval) | **Oracle Edison Award** | Academic Star, IIIT Bangalore | Top Performer, Abzooba